

SWAGARIKA JAHARLAL GIRI

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BACKGROUND

Ph.D. researcher in Computer Science with **7+** years of experience in machine learning and NLP for scientific and biological applications, complemented by **3** years of industry experience as a full-stack developer **Publicis Sapient**. Completed research internships at **Qualcomm (Summer 2024)** and **SanDisk (Summer 2026)**. Author of **11+** publications, with **1** patent filed and **75+** citations. Research focuses on protein function prediction using **LLMs**, **RAG**, **Agentic systems**, and **GraphRAG**.

EDUCATION

Ph.D. in Computer Science, Purdue University, USA Jan 2022 – Present

CGPA: **3.83**/4.00

Research focus: Bridging Interpretability and Accuracy in Protein Function Prediction Using LLMs.

M.Tech in Computer Science and Engineering, IIT Patna, India 2017 – 2019

CGPA: **9.31**/10

EXPERIENCE

Research Intern, Sandisk May 2026 – Aug 2026

Developed an Agentic AI SmartFab platform using LLMs, RAG, Vision LLMs, and Video LLMs for semantic search, question answering, multimodal meeting summarization, interactive dashboards, and 3D fab visualization.

Research Intern, Qualcomm May 2024 – Aug 2024

Developed a constraint-aware, LLM-driven travel planning system leveraging structured tool-calling across flight, routing, scheduling, and validation APIs, reducing planning latency by 30% and leading to a **patent filing**.

PhD Researcher, Purdue University, Kihara Lab Jan 2022 – Present

- Queryome: Co-developed an **agentic system** that orchestrates multi-step querying, retrieval, and relevance assessment over large-scale biomedical corpora, enabling protein-centric evidence discovery and question answering.
- GO2Sum: **LLM-based summarizer** translating Gene Ontology (GO) term sets into human-readable protein function summaries, outperforming SciFive and template-based baselines (*NPJ Systems Biology and Applications*, 2024).
- Lit-PFP: **A RAG-based** literature-driven protein function prediction framework combining BM25 sparse retrieval, FAISS dense embeddings, and hybrid RRF re-ranking with LLM-based semantic filtering — further refined through reinforcement learning to maximize recall of experimentally validated GO annotations.
- GraphRAG-PFP: Developed a **GraphRAG** framework that transforms unstructured corpora into hierarchical knowledge graphs using LLM-based entity extraction and Leiden community detection, enabling structured global reasoning beyond traditional vector retrieval.

Full-Stack Developer, Publicis Sapient Jul 2019 – Nov 2021

Contributed as a **full-stack developer** across large-scale data and e-commerce systems, building Spark- and Scala-based data transfer pipelines for CJ Affiliate while developing and deploying production-grade React applications, including a Shopify app, on AWS with automated CI/CD workflows.

Research Assistant, IIT Patna, AI-ML-NLP Lab Jun 2017 – Jun 2019

Developed **two** deep multimodal learning systems integrating heterogeneous biological data for functional prediction, **later published in IEEE journals and conferences**, and deployed as scalable AWS-based web applications.

SELECTED PUBLICATIONS

- **Giri SJ**, Ibtehaz N, Kihara D. *GO2Sum: Generating human-readable protein function summaries from GO terms*. NPJ Systems Biology and Applications, 2024.
- **Giri SJ**, Dutta P, Halani P, Saha S. *MultiPredGO: Deep multi-modal protein function prediction by combining structure, sequence, and interaction information*. IEEE Journal of Biomedical and Health Informatics, 2020.
- **Giri SJ**, Saha S. *Multi-View Gene Clustering using Gene Ontology and Expression-based Similarities*. IEEE Congress on Evolutionary Computation (CEC), 2020.

TECHNICAL SKILLS

ML/NLP: PyTorch, Transformers, LLMs, GraphRAG, FAISS, RAG, BM25

Languages: Python, Java, C++, SQL, JavaScript

Frameworks: React, Flask, Spark, AWS

Domains: Computational Biology, Protein Function Prediction, Multimodal Learning, Natural Language Processing

ACADEMIC SERVICES AND RECENT AWARDS

Reviewer: ICML, ICLR, IEEE/ACM TCBB

Maintainer of Web servers: PFP, Phylo-PFP, ESG, GO2Sum, NaviGO, 3D-Surfer (Kihara Lab)

Received the *Graduate Teaching Award* from the Purdue University Department of Computer Science (Fall 2025).

Received *Partnering for Client Impact Award*, Publicis Sapient (2019)